

stats

Descriptive

Inferential

Descriptive

- (a) measures of central tendency
- (b) measures of dispersion
- (c) measures of shape
- (d) measures of position
- (e) data visualization
- (f) measures of relationship

measures of central tendency

(i) mean =

10, 20, 30, 40, 50

pop. $\mu = \frac{150}{5} = 30$

sample $\bar{X}$

② median

$$[10, 20, \underline{30}, 40, 50, 60]$$

$$\text{median} = \frac{30 + 40}{2} = 35$$

③ mode

$$[2, 3, 4, 2, 5, 4, 2]$$

$$\text{mode} = 2$$

Ⓐ Unimodal

Ⓑ bimodal

Ⓒ multimodal

④ weight mean $\Rightarrow$

$$[10, 20, 30, 40]$$

$$\Rightarrow \frac{10 \times 0.4 + 20 \times 0.2 + 30 \times 0.6 + 40 \times 1}{4}$$

$$\Rightarrow \frac{4 + 4 + 18 + 40}{4} = 16.5$$

measures of dispersion

I) Range = max-min

$$[10, 20, 15, 16, 1, 2, 4, 8]$$

$$= 20 - 1 = 19$$

II) Variance - spread of data

$$\text{pop } \sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

$$\text{sample } s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$$

x_i	$\bar{x}$	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$	$\sum (x_i - \bar{x})^2$
2	5.6	-3.6	12.96	<u>33.36</u>
4		-1.6	2.56	
5		-0.6	0.36	
6		0.4	0.16	
8		2.4	5.76	
9		3.4	11.56	

$$\Rightarrow \frac{33.36}{(6-1)} = \frac{33.36}{5} \Rightarrow \boxed{6.67}$$

III) standard deviation $\rightarrow$

$$\text{pop } \sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{n}}$$

$$\text{sample } s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

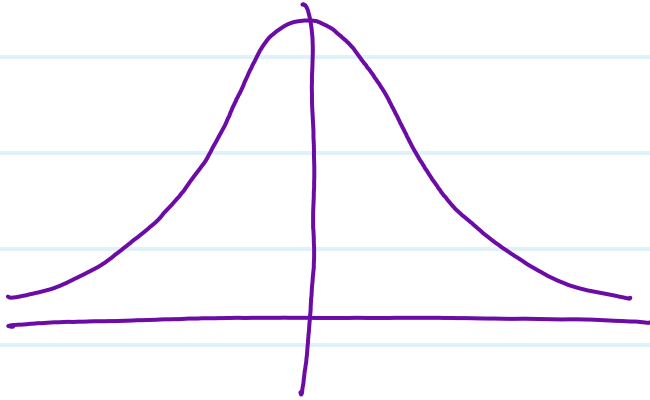
$$s = \sqrt{6.67}$$

$$= 2.58$$

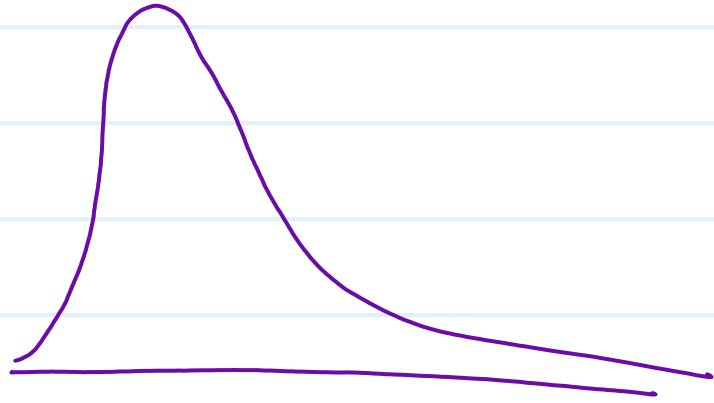


measures of shape

I) skewness

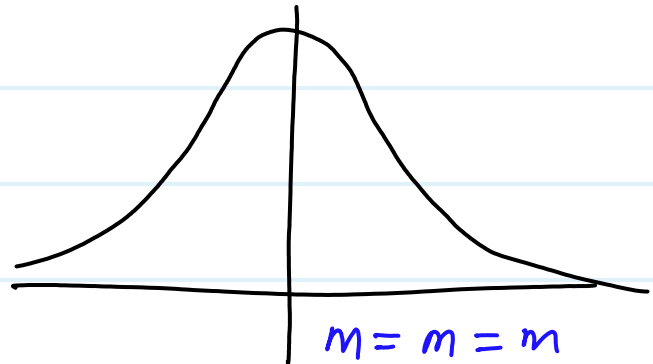


Symmetric



Asymmetric

① zero skewness



$m = m = m$

② Right skewed



$m > m > m$

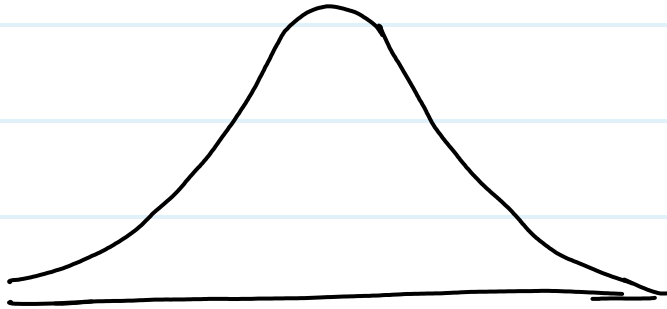
③ Left skewed



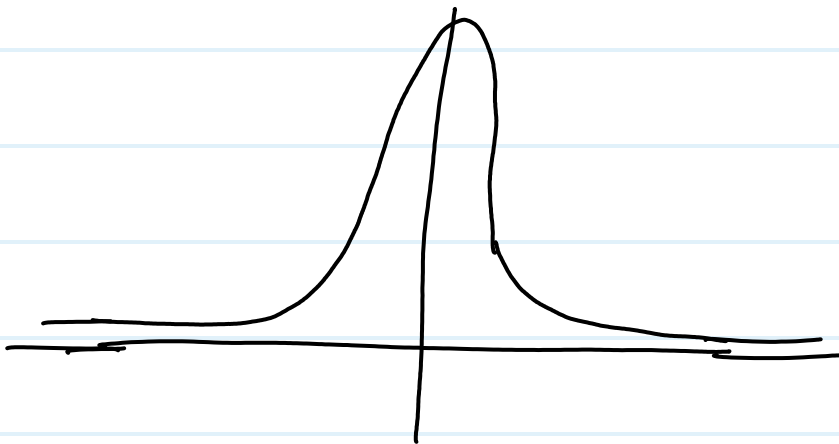
$m < m < m$

II) kurtosis

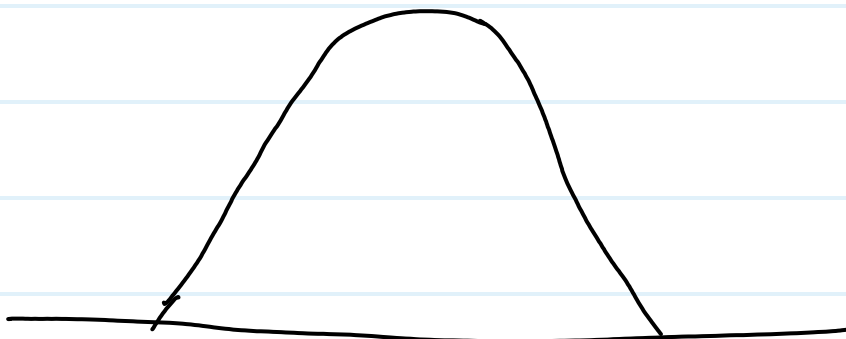
① mesokurtosis $K=3$



② Leptokurtosis $K > 3$



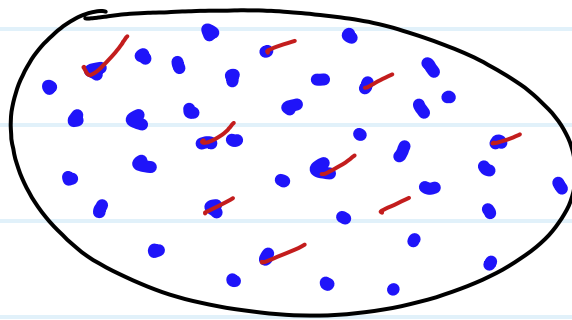
③ platykurtosis $K < 3$



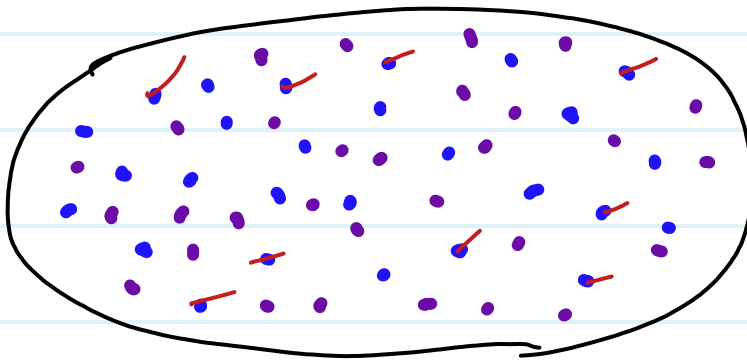
Population (N)

Sample (n)

① Simple random sampling



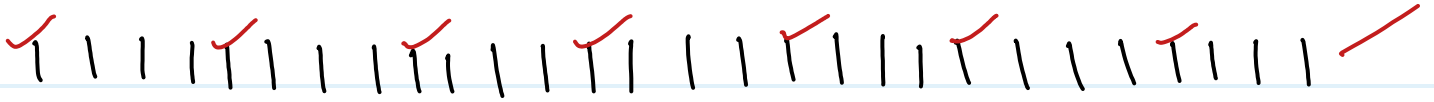
② stratified sampling



1000
600
—
↑

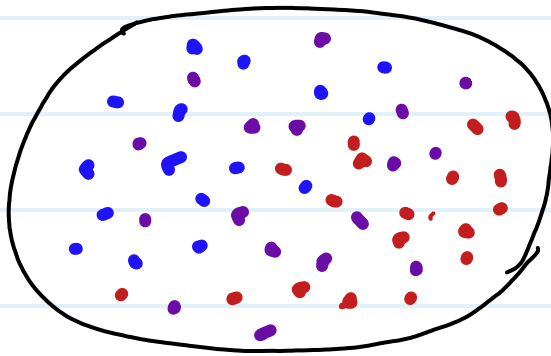
educated female →

③ systematic sampling



n^{th}

④ cluster sampling -



⑤ convenience sample